

University of the Armed Forces

DEPARTMENT OF COMPUTER SCIENCE

Object-Oriented Programming

Homework 3-Software Requirements Specification

Group N°1

Members:
ISAAC SEBASTIAN ACURIO CATOTA
ARMAS GUAÑO DEMIAN ROLANDO
ENRIQUE MATEO AYALA LAGUNAS

Edison Lascano PhD
NRC: 36862

Software Requirements Specification (SRS) and Architectural Design: Liquor Track Project

1. Product Perspective and Operational Context

This document establishes the formal software requirements specification and architectural guidelines for "Liquor Track," a software engineering project designed to modernize and centralize the daily operations of a commercial liquor store. Currently, the store's operational management faces severe limitations due to manual processes. Inventory control is done visually, which delays the identification of stockouts; the management of credit accounts lacks structured tracking; and the application of discounts relies entirely on the cashier's memory, generating financial vulnerabilities and a poor understanding of cash flow.

The conceptualization of Liquor Track responds directly to the need for an integrated digital ecosystem that unifies the Point of Sale (POS), inventory management, supplier control, and cash reconciliation. This specification adopts standard recommended practices to ensure that requirements are unambiguous, verifiable, and traceable throughout the development lifecycle.

The main purpose of this document is to establish a solid architectural foundation for the development team (Isaac Acurio, Demian Armas, and Enrique Ayala) and to ensure that the implemented technology aligns perfectly with the store's business goals. The system aims to mitigate human errors, provide real-time business intelligence through sales reports, and ensure compliance with strict local laws.

2. Legal, Regulatory, and Business Constraints

Designing and implementing a software system for selling liquor in Ecuador requires strict alignment with legal and tax regulations. These constraints shape the core business logic of the application and impose validation rules that the system cannot bypass under any circumstances.

2.1 Age Control and Penal Regulation

The sale of alcoholic beverages is heavily regulated to protect minors. In Ecuador, the law strictly prohibits the sale and distribution of alcohol to anyone under 18 years of age. The Comprehensive Organic Penal Code (COIP), in its Article 396, classifies this action as a fourth-class contravention, imposing penalties that include 15 to 30 days of prison for the seller, in addition to financial fines and the potential definitive closure of the commercial establishment.

Consequently, the software must incorporate an unavoidable blocking mechanism (hard stop) in the POS interface. Before processing any payment containing items categorized as "Alcoholic Beverages," the system will require an explicit confirmation from the cashier, certifying that they have verified the buyer's ID document and that the customer is of legal age. Omitting this step will make it transactionally impossible to complete the sale.

2.2 SRI Electronic Invoicing Regulations

The Ecuadorian tax environment imposes technical requirements for issuing sales receipts. Recent

reforms by the Internal Revenue Service (SRI), driven by Executive Decree 586, have drastically modified the thresholds for anonymous billing.

Historically, businesses could issue invoices to a generic "Consumidor Final" (Final Consumer) for amounts up to $200 USD. However, current legislation dictates that any commercial transaction exceeding $50 USD strictly requires the full identification of the buyer using their ID number (Cédula), RUC, or passport. Liquor Track will address this restriction by operating as a tax sentinel: the system will intercept any transaction totaling more than $50 USD, block the "Consumidor Final" option, and force the entry of the customer's tax data before allowing the payment to proceed.

3. Software Architecture and Object-Oriented Design

To ensure the maintainability, scalability, and robustness of the system, the architecture of Liquor Track will be built upon Object-Oriented Programming (OOP) principles and the Model-View-Controller (MVC) architectural pattern. The MVC pattern will maintain a strict separation between data presentation, user interaction, and the underlying business logic.

3.1 Behavioral Design Patterns

The dynamic behavior of the system will be modeled using the following standardized design patterns to elegantly resolve the variability of business rules:

- Strategy Pattern: This pattern will be implemented in the pricing engine to abstract the logic of promotions and discounts. The POS will delegate the final price calculation to a DiscountStrategy interface, which can dynamically instantiate percentage discounts or wholesale prices at runtime, avoiding fragile conditional statements (if/else chains) in the code.
- Observer Pattern: This will guarantee a reactive architecture for inventory monitoring. When a product's stock level drops below its parametric minimum threshold, the system will automatically notify the owner without tightly coupling the sales module to the notification module.
- State Pattern: The lifecycle of purchase orders will be orchestrated through this pattern. A purchase order will transition through encapsulated states such as Pending, Received, or Canceled. The inventory stock in the database will only increase when the order executes a legitimate transition to the Received state.

3.2 Concurrency and Optimistic Locking

Point of Sale systems are inherently subject to concurrency issues. A classic race condition occurs when multiple cashiers attempt to sell the last available unit of a highly popular product at the exact same time. To safeguard data integrity without causing performance bottlenecks, the system's persistence layer will utilize Optimistic Locking. This technique detects conflicts during the database commit phase, preventing negative inventory while maintaining fast response times.

4. Structured Requirements Specification

The system requirements are broken down into functional and non-functional attributes, structured in a tabular format to facilitate tracking and testing.

4.1 Functional Requirements (FR)

ID	Domain	Expected Feature / Behavior	OOP Rule / Pattern
FR-01	Security	The system shall require encrypted authentication and differentiate access levels between Owner, Admin, and Cashier roles.	Role-Based Access (RBAC)
FR-02	Catalog	The system shall allow users to register, view, edit, and deactivate product entities (codes, costs, prices, stock limits).	Domain Entities
FR-03	Inventory	The system shall allow the owner to manage product stock, recording stock entries and exits and displaying current quantities. Every completed sale shall automatically deduct	Domain Entities / Optimistic Locking

		stock.	
FR-04	Alerts	The system shall emit automated alerts to the owner when a product's inventory drops to or below its safety threshold.	Observer Pattern
FR-05	Supply	The system shall manage supplier information and pending orders. Inventory will only increase upon confirming the physical receipt of the order.	State Pattern
FR-06	POS	The system shall provide an interface to add products to a cart, calculating subtotals and totals in real-time.	MVC Controller
FR-07	Sales	The system shall record every sale under the active cashier and allow the owner to review the sales made by each employee.	Employee Context in Transaction

FR-08	Financial	The system shall manage customer credit accounts, tracking total debt and recording partial payments safely. The owner and the cashier shall be able to record payments and view credit sales, and the owner shall be able to view each customer's outstanding balance.	Immutable History
FR-09	Closing	The system shall allow recording daily expenses and shall generate a daily cash-closure report with total cash and expenses to reconcile the day's income.	Reconciliation
FR-10	Pricing	The system shall automatically calculate default selling prices by applying a 20% profit margin on the product's purchase cost, and the owner shall be able to configure this margin	Business Rule

		per product.	
FR-11	POS	The system shall automatically and dynamically apply active promotions and discounts when time or volume conditions are met, preventing manual charging errors.	Strategy Pattern
FR-12	Reports	The system shall track historical sales data to generate a best-selling products report that supports purchasing decisions.	Data Aggregation
FR-13	Compliance	Age Check: The POS must display a mandatory dialog requiring age confirmation before completing alcohol sales.	COIP Art. 396

FR-14	Compliance	Tax Limit: The system must block "Consumidor Final" billing if the total exceeds $50 USD, forcing tax data entry (ID number, RUC, or passport).	SRI Decree 586

4.2 Non-Functional Requirements (NFR)

To govern performance and software quality, the following measurable parameters are based on the ISO/IEC 25010 classification. The thresholds are verified through the tests described for each requirement and can be adjusted by the team.

ID	Quality Attribute	Restrictive Condition and Performance Threshold
NFR-01	Performance	Catalog searches and checkout total calculations shall respond in 2.0 seconds or less for at least 95% of requests under standard load (up to 3 concurrent POS terminals).
NFR-02	Usability	A new cashier shall be able to complete a standard sale in no more than 5 clicks after adding the products, and at least 90% of new cashiers shall complete a sale without assistance after 30 minutes of training.

NFR-03	Fault Tolerance	The system shall use Optimistic Locking to detect 100% of conflicting concurrent inventory updates, resulting in 0 negative stock records, and at least 99% of valid transactions shall complete successfully on the first attempt under standard load.
NFR-04	Maintainability	The persistence layer shall be isolated through modular segregation so that replacing the database requires changes in no more than 10% of the source files, and business-logic unit tests shall reach at least 80% code coverage.
NFR-05	Availability	The system shall be available at least 99% of the store's opening hours per month, with automated daily database backups completed successfully 100% of days, a maximum data loss (RPO) of 24 hours, and a maximum recovery time (RTO) of 4 hours.

5. Requirements Traceability Matrix (RTM) and Testing Strategy

To demonstrate the thoroughness of the engineering process, the software design includes a Requirements Traceability Matrix (RTM). This instrument creates clear links between the original User Stories (US) from the Product Backlog and the technical testing strategy. Testing will be executed using JUnit 5,

verifying precise data flows and isolating components using mocking techniques in Java.

User Story ID	Original Backlog Need	Req ID	Descriptive Verification Method (JUnit 5)
US001	The owner needs to manage product stock (entries, exits, and current quantities).	FR-03	verifyStockQuantityUpdatesOnEntryAndExit()
US002	The owner requires early low-stock warnings.	FR-04	verifyObserverNotifiedWhenStockDropsBelowThreshold()
US003	The admin will manage suppliers and track pending orders.	FR-05	testStateTransitionToReceivedIncreasesGlobalInventory()
US004	Sales must be logged individually	FR-07	assertTransactionRecordsEmployeeContextIdAccurately()

	by employee.		
US005	The owner needs to manage credit payments.	FR-08	assertRemainingBalanceDecreasesCorrectlyOnCreditPayment()
US006	Generate a daily cash closure with total cash and expenses.	FR-09	assertCashClosingReportsTotalCashAndExpenses()
US007	The system must apply a 20% profit margin to products.	FR-10	testDefaultSellingPriceAppliesConfiguredProfitMargin()
US008	The system must automatically apply active discounts and promotions.	FR-11	testDiscountStrategyInjectsCorrectPercentageIntoTotal()

US009	A report distilling fast-selling products is needed.	FR-12	assertBestSellingAlgorithmSortsQuantitiesInDescendingOrder()
US-Legal-01	Cashiers must certify customer age for alcohol.	FR-13	assertTransactionThrowsExceptionIfAgeVerificationFlagsFalse()
US-Legal-02	SRI tax limit for anonymous receipts.	FR-14	testInvoiceBlocksFinalConsumerEntityIfTotalExceeds50USD()
US-System	Prevent concurrent sales errors on multiple devices.	NFR-03	testOptimisticLockExceptionThrownOnConcurrentInventoryUpdate()

6. Final Considerations

The design of Liquor Track provides a highly robust technical response to the organizational flaws of the previous business model. By articulating complex methodologies, such as Object-Oriented design patterns and the MVC architecture, the team guarantees optimal long-term maintainability.

Beyond the strictly technological scope, this report emphasizes that the software does not operate in a regulatory vacuum. Embedding the strict rules stipulated by the Ecuadorian penal code (COIP) and the Internal Revenue Service (SRI) transforms this tool into a legal and financial risk mitigation agent for the store. This specification provides the definitive blueprint that will govern the coding, auditing, and progressive implementation process undertaken by the engineering team.

Bibliography

1. Drinking Age in Ecuador (2026): Legal Purchase & Consumption Age, https://drinkingagein.com/ecuador/
2. Cárcel por vender licor a menores - Derecho Ecuador, https://derechoecuador.com/carcel-por-vender-licor-a-menores/
3. Registro Oficial No.475- Jueves 25 de Abril de 2019, https://derechoecuador.com/registro-oficial-no475-jueves-25-de-abril-de-2019/
4. Tope consumidor final 2026 Ecuador: USD 50 | Blog AZUR, https://azur.com.ec/blog/monto-maximo-para-facturar-a-consumidor-final-en-ecuador-2026-lo-que-debes-saber
5. Consultar los principales cambios en el decreto 586 - Contífico, https://contifico.portaldeclientes.siigo.ec/basedeconocimiento/consultar-principales-cambios-decreto-586/
6. Strategy Pattern - DEV Community, https://dev.to/stevsharp/strategy-pattern-h9l
7. Design Patterns Elements of Reusable Object-Oriented Software, https://www.javier8a.com/itc/bd1/articulo.pdf
8. Optimistic vs Pessimistic Locking: The System Design Interview Guide, https://spacecomplexity.ai/blog/optimistic-vs-pessimistic-locking-system-design
9. ISO/IEC 25010:2023 Guide to Software Product Quality Standards, https://blog.pacificcert.com/iso-25010-software-product-quality-model/